Available online at www.sciencedirect.com

ScienceDirect

Current Opinion in Electrochemistry

Review Article

The emergence of automation in electrochemistry

Michael A. Pence[1,2], Gavin Hazen[1,2] and Joaquín Rodríguez-López[1,2]

Automated electrochemistry is emerging as a powerful tool to accelerate discoveries in important fields such as energy storage, catalysis, electrosynthesis, and electrochemical sensing, among others. Automated electrochemistry platforms (AEPs) offer the ability to increase the throughput of electrochemical experiments, reduce the workload of manual experimenters, and offer opportunities for enhanced reproducibility and transferability of experimental protocols. This review explores the design, capabilities, and applications of AEPs, highlighting platforms constructed with various fluid handling, motion control, and signal multiplexing capabilities for a variety of applications. Further, we highlight platforms that integrate automated synthesis and electrochemical characterization and we discuss the role of Bayesian optimization in autonomous experiments, enabling both performance optimization and mechanistic insight. Finally, we offer an outlook on the potential future directions of automated electrochemistry.

Addresses
[1] Beckman Institute for Advanced Science and Technology, University of Illinois Urbana-Champaign, Urbana, IL 61801, United States
[2] Department of Chemistry, University of Illinois Urbana-Champaign, Urbana, IL 61801, United States

Corresponding author: Rodríguez-López, Joaquín (joaquinr@illinois.edu)

Current Opinion in Electrochemistry 2025, **51**:101679

This review comes from a themed issue on **Innovative Methods (2025)**

Edited by **Paul W. Bohn** and **Alan O'Riordan**

For complete overview about the section, refer Innovative Methods (2025)

Available online 8 March 2025

https://doi.org/10.1016/j.coelec.2025.101679

2451-9103/© 2025 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

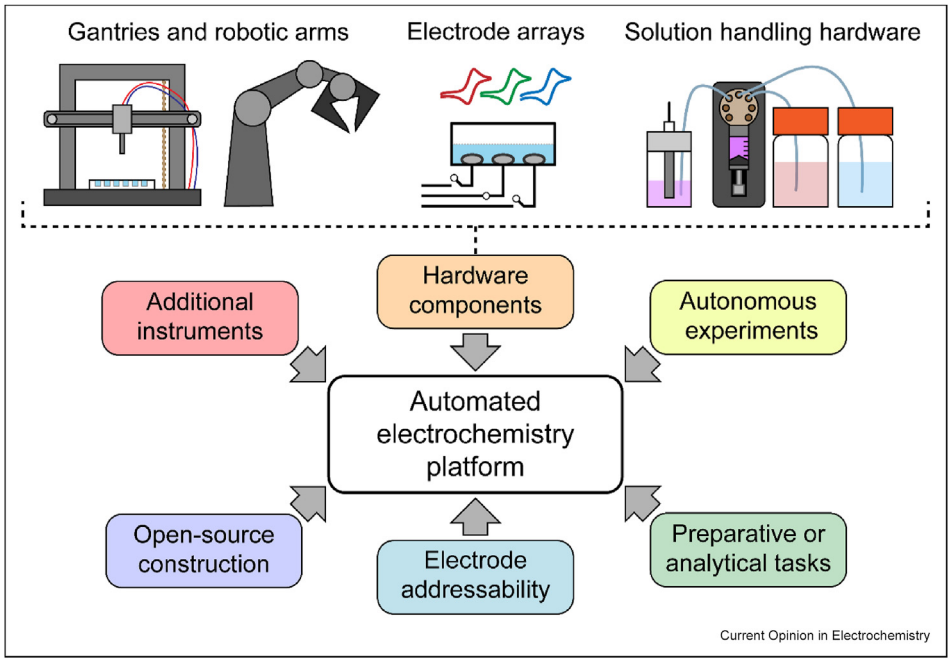
Introduction

Electrochemistry offers effective solutions to pressing environmental and climate challenges, with electrochemical technologies becoming increasingly prevalent in many fields such as energy storage and conversion, sustainable synthesis, carbon capture, water desalination, pollution remediation, and environmental sensing [1,2]. As the climate crisis intensifies, the need for rapid discovery in electrochemistry has never been more urgent. Automation provides a powerful pathway to accelerated discovery, offering increased throughput, consistent reproducibility, and the ability to conduct passive, high-dimensional experimental campaigns [3–7]. When paired with machine learning (ML), automation helps minimize the work needed for extensive experimental campaigns on unknown systems with many parameters [8]. These strategies enable researchers to rapidly obtain a wealth of data that can complement—rather than replace—their chemical intuition, empowering new discoveries.

Despite automation gaining mainstream acceptance in the physical sciences, it has only recently emerged in the electrochemistry community, with a rapid uptick in relevant works published in the past two years. However, the concept of automation in electrochemistry is not entirely new; work by Cullen et al. demonstrated fully automated solution handling and polarography as early as 1973 [9]. There is a rich history of related efforts in high throughput electrochemistry, some even finding extensive commercial success in the form of electrochemiluminescence (ECL) based screening platforms for biomedical and diagnostic applications [10–12]. However, we note a nuanced difference in the goals of automation and high-throughput experiments, where the former seeks to minimize human intervention and the latter seeks to maximize experimental efficiency, although these goals are not mutually exclusive. Despite early demonstrations of automated electrochemistry and related efforts, adoption has been slow, due to the complexity of designing and implementing automated electrochemistry platforms (AEPs). As such, this work is meant to familiarize the reader with existing AEPs, empowering them to build their own. By exploring a diverse range of AEPs of varying complexity, capabilities, and applications, this work aims to inspire researchers to embrace automation and integrate these powerful tools into their own research efforts.

Building an automated electrochemistry platform (AEP) can be a challenging endeavor, requiring meticulous planning and foresight to successfully automate the desired electrochemical experiments. Figure 1 highlights key considerations for researchers designing an AEP. For example: Will the platform be used for preparative electrochemistry or for electroanalysis? Will it integrate additional instruments or specialized

Figure 1



Schematic of major influencing factors for automated electrochemistry platform (AEP) design, including examples of common hardware components for fluid handling, motion control, and signal multiplexing.

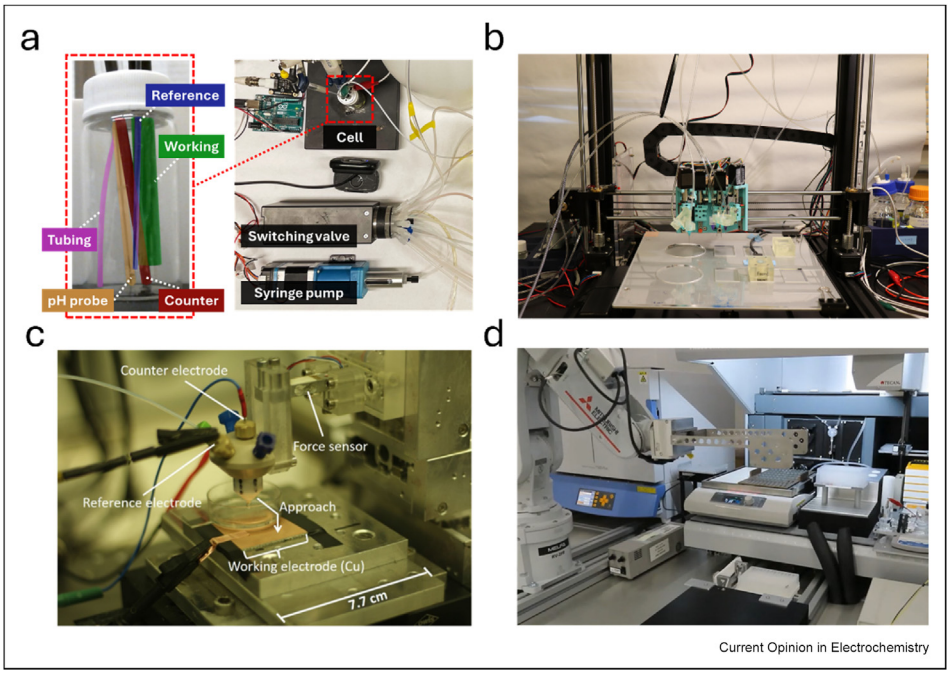
hardware? Should it enable autonomous experiments? Is an open-source design available? These questions must be carefully addressed during the design process and can often be overwhelming for those new to automation. Additionally, we note that it is important to consider experimental conditions that could affect the design of the AEP, such as the need for experiments to be in air free environments [13,14] or to perform low current measurements [15].

Hardware components for automated electrochemistry

Common AEP hardware beyond electrochemical instrumentation performs three primary tasks: fluid handling, motion control, and signal multiplexing. Fluid handling allows for dynamic preparation of solutions in either batch or flow configurations, often using either automated pump-valve setups or pipetting robots [14,16,19–25]. Common pump types include peristaltic pumps for batch reactors, pressure displacement pumps for flow reactors, and syringe pumps for multiple types of reactors [26–29]. AEPs utilizing solution handling are ideal for studies requiring precise control of reagent concentration and are commonly used in applications in molecular electrochemistry, where the electrode–electrolyte interface is not a primary concern. For example, our eLab platform (Figure 2a) interfaced a conventional 3-electrode cell with automated solution handling hardware to study the pH dependence of oxoammonium-catalyzed alcohol electro-oxidation [16]. We used a syringe pump and 16-channel switching valve, addressing our needs for precise and accurate titration experiments and multi-substrate screening campaigns respectively. Automated fluid handling enables real-time tuning of reagent concentrations, which is often essential for autonomous studies where solution parameters are dynamically adjusted based on algorithmic suggestions [14,19,21,23,30].

Motion control provides a means to mechanically manipulate the experimental setup, and is typically employed using gantries, robotic arms, or scanning probe type setups. Gantries are an accessible and cost-effective option. Figure 2b show the Electrolab, a platform from our group that utilized an XYZ gantry taken from a low-cost 3D printer for motion control, moving a multi-nozzle head to a desired electrochemical cell to perform fluidic operations [15]. In principle the gantry design of the Electrolab could enable the experimenter to dispense solutions on multiple, independent electrochemical cells for both reproducibility testing and exploratory work. XYZ gantries are commonly used to move electrodes between solutions in well-plate formats, a setup applied to problems in electrocatalysis [31], electrochromics [19], sensor development [32,33], and other fields [34–37]. Zeng et al. designed a gantry system to study CO_2 electro-reduction on tetrapyrrole-

Figure 2



Automated electrochemistry platforms (AEPs) with various hardware configurations. **(a)** The eLab platform, consisting of a conventional three-electrode cell and pump-valve type solution handling for molecular electrocatalysis studies. Figure reproduced with permission from Pence et al. [16] Copyright 2023, Royal Society of Chemistry. **(b)** The Electrolab platform that combines fluidic handling, motion control, and electrode arrays for redox-flow battery studies. Figure reproduced with permission from Oh et al. [15] Copyright 2023, Elsevier. **(c)** A scanning droplet cell platform for Li plating studies. Figure reproduced with permission from Dieckhöfer et al. [17] Copyright 2021, Wiley-VCH. **(d)** A dual robotic arm platform for electrocatalysis studies. Figure reproduced with permission from Kodera et al. [18] Copyright 2023, Royal Society of Chemistry.

modified electrodes, uniquely using the gantry to seal a cell over each electrode for electrolysis and analyzing the products by gas chromatography downstream [38].

A variation of the XYZ gantry is the scanning droplet cell (SDC), an emerging theme in AEPs that originates from scanning electrochemical probe microscopy techniques [39,40]. The SDC system (Figure 2c) operates by confining the electrolyte, reference, and counter electrodes in a miniaturized electrochemical cell (the SDC head), which is mounted on an XYZ positioner and pressed onto the sample surface, enabling spatially localized electrochemical reactions [17]. The electrolyte within the SDC head can be adjusted using automated solution handling, which has enabled studies of materials for electrocatalysis and batteries [17,41–46]. The SDC approach is powerful for performing automated experiments across large heterogenous electrodes, but the SDC hardware must be custom designed [47,48].

While gantries are powerful, robotic arms, though costlier and more complex, offer greater flexibility and can perform advanced, multi-step electrochemical tasks that require human-like finnese [18,20,49,50]. For example, AEPs based on robotic arms have been used for automatic electrode polishing in between electrochemical experiments [20,51]. Figure 2d highlights the dual robotic arm platform developed by Kodera et al. for high-throughput screening of Co-Mn-Fe-Ni composite-oxide electrocatalysts. This AEP utilized one robotic arm for sample manipulation and the other for solution pipetting. The system automated a multi-step and multi-instrument autonomous workflow that included electrode preparation, calcination, electrochemical characterization, and product identification— an automated routine that necessitates the flexible mobility afforded by robotic arms.

Augmenting an AEP with hardware such as electrode arrays, multiplexers, and multichannel potentiostats can increase experimental throughput. Additionally, using multiple electrodes enables studies of electrode modification and can circumvent the effects of electrode fouling [19,52–57]. Serially addressed electrode arrays use a multiplexer to select a desired electrode among many present in a common solution [55,58]. Alternatively, counter and reference electrodes can be mechanically transported to a common cell with a desired working electrode, a strategy used in well-plate arrays [34]. Parallel multielectrode setups achieve the highest level of throughput, by simultaneously

applying potential to multiple electrodes and recording the resulting current for each electrode [34,59]. The Legion platform exemplifies this setup, enabling the simultaneous operation of 96 electrodes for both analytical and preparative applications [60–62]. A 'wireless' approach to parallel electrochemistry was recently demonstrated using microfabricated photovoltaic devices, known as SPECS, to drive electrosynthetic reactions [63].

In summary, we outline the following key hardware motifs and motivations for their use.

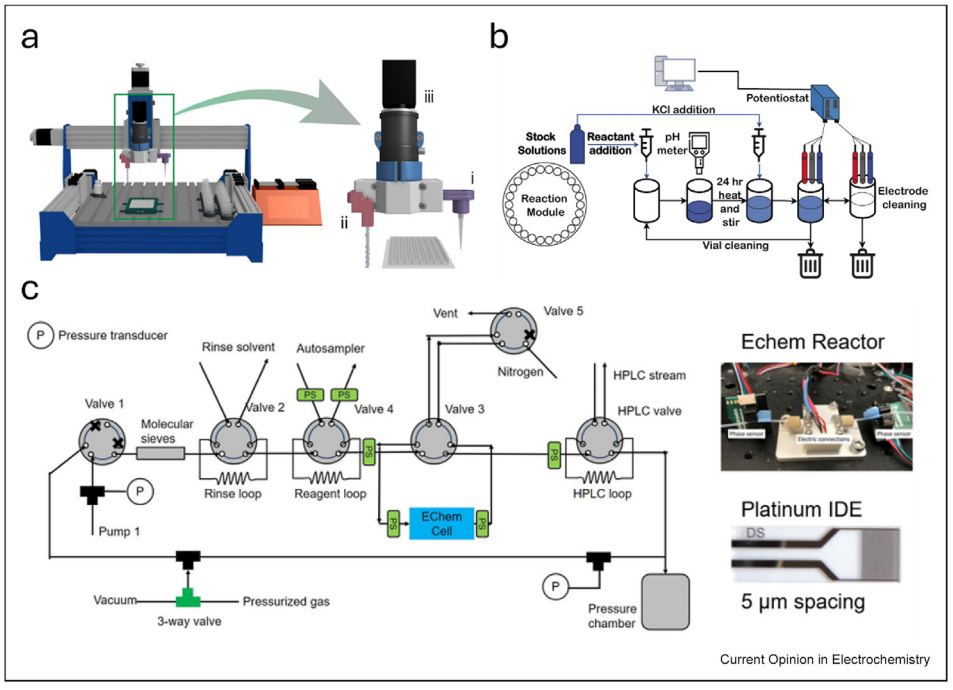
- Solution handling hardware provides in-demand preparation of solutions, enabling combinatorial searches, mechanistic investigation, and titration-type studies.
- Motion control hardware provides human-like experimental flexibility but with increased electromechanical complexity, ideal for manipulating electrodes, switching electrochemical cells, and precision tasks like electrode polishing.
- Electrode arrays can facilitate the study of electrode modification, mitigate fouling effects, and enable parallel measurements for high-throughput experimentation

Coupling automated synthesis and characterization

While many platforms discussed in this work were designed for automated electrochemical analysis, AEPs can also be used for preparative electrochemistry. Platforms that combine both synthesis and analysis significantly enhance the functionality of AEPs, enabling integration of the platform into the design–make–test–analyze cycle when combined with ML or human guided decision making [64]. Figure 3 shows a variety of platforms that combine synthesis and characterization steps for these type of workflows.

Platforms that integrate preparative and analytical electrochemistry into a single workflow have been demonstrated primarily through electrodeposition-based materials synthesis, followed by electrochemical screening. Figure 3a highlights PANDA, an AEP designed for high-throughput electrochromic polymer deposition and characterization that utilizes a gantry-based system to serially address a 96-electrode well plate. PEDOT:PSS films were electrodeposited and subsequently screened spectro-electrochemically, with a camera capturing electrochromic color changes under applied potential. PANDA incorporated Bayesian optimization to enable fully autonomous tuning of

Figure 3



Example of AEPs that perform preparative and analytical operations. (**a**) The PANDA platform that electrochemically prepares and analyzes electrochromic materials. Figure reproduced with permission from Quinn et al. [19] Copyright 2024, Royal Society of Chemistry. (**b**) The ECARUS platform performs inorganic synthesis and downstream electrochemical analysis of POMs. Figure reproduced with permission from Laws et al. [65] Copyright 2023, Wiley-VCH (**c**) An automated flow system uses interdigitated array eleectrodes to perform preparative electrosynthetic cross-coupling reactions with downstream chromatography for product analysis. Figure reproduced with permission from Zahrt et al. [22] Copyright 2022, American Chemical Society.

electrodeposition parameters to achieve the best electrochromic performance. Gaddam et al. used an automated electrode array workflow to speed up discovery of electrodeposited electrocatalysts. In this array-based workflow, Au modified electrodes were prepared by electrodeposition and screened for electrocatalytic activity for the glycerol oxidation reaction [56]. We note here that both platforms require the use of electrode arrays, as the permanent modification from electrodeposition precludes reusing electrodes for additional experiments.

The integration of molecular and materials synthesis with downstream electrochemical characterization is a new and exciting direction in the field. The ability to synthesize desired species on demand together with automated characterization enables fully unsupervised exploration beyond a predefined analyte scope. Figure 3b shows ECARUS, an AEP that uses CV to characterize automatically synthesized polyoxometallates (POM) [65]. ECARUS integrates an inorganic synthesis robot [66] with an AEP, enabling fully autonomous exploration of POM clusters with the ultimate goal of developing improved pseudo-capacitor materials. Pablo-García et al. coupled a synthesis robot with an open-source potentiostat to prepare and electrochemically characterize organometallic complexes [67]. Further, solid-state materials synthesis workflows (beyond electrodeposition) have been coupled with AEPs, where electrocatalyst precursors were drop-cast onto electrodes and thermally processed before being automatically characterized for electrocatalytic activity [18,31]. The coupling of synthetic platforms to electrochemical characterization has been primarily limited to single-step synthesis, but state-of-the-art synthesis robots will greatly expand the accessible chemical space [4,5,7,68,69].

Beyond electrochemical characterization, AEPs have been used to perform electrosynthesis reactions, where products are analyzed downstream by additional analytical instrumentation. Notably, the platform developed by Zahrt et al. in Figure 3c used an electrochemical flow cell for discovery of radical−radical cross-coupling reactions, with products being automatically analyzed by downstream HPLC [22]. The electrochemical flow cell used an interdigitated array electrode to perform paired electrolysis, using the short inter-electrode diffusion lengths (5 μm) to ensure successful coupling reactions. Beyond reaction discovery, Liang et al. demonstrated an automated flow platform for autonomous process optimization of tetrabenzylthiuram disulfide electrosynthesis [23]. AEPs for preparative electrochemistry are likely to become common place given the resurgence of interest in organic electrosynthesis [70].

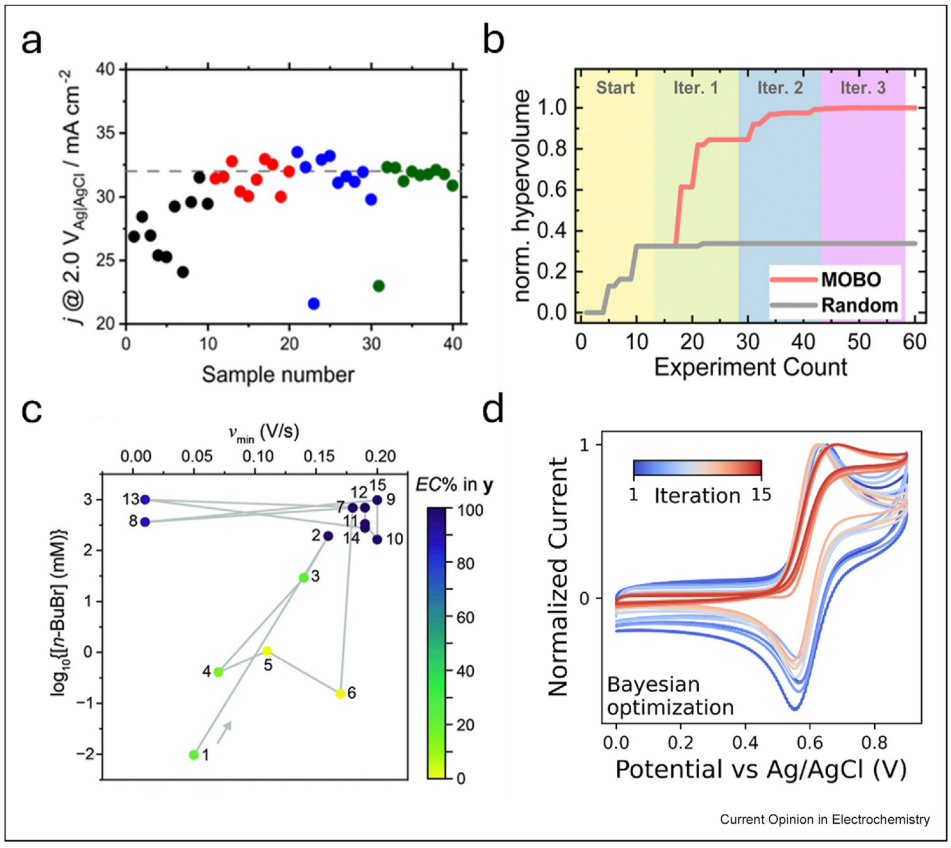
In summary, the coupling of synthesis and characterization robots represents an exciting new direction, where planning the design of each component with modularity in mind is good practice.

Autonomous experiments driven by machine learning

Integrating AEPs with machine learning (ML) for decision making enables fully autonomous experimental campaigns that span expansive multi-dimensional parameter spaces. Often, human intuition alone is not sufficient to navigate such complex problems, especially when facing cost and budget constraints. Algorithm guided experimentation offers a solution to these challenges, utilizing data driven approaches to make decisions during experimental campaigns. While there are many algorithms to autonomously drive experiments [71], Bayesian optimization is one of the most utilized, performing optimization across black-box functions, often with continuous input variables [72]. Bayesian optimization uses previously acquired data to construct a surrogate model and identify parameters likely to enhance a target metric [73]. Parameters are chosen from a defined domain by an acquisition function, which often requires the selection of an exploration-exploitation hyperparameter. Tuning the amount of exploration or exploitation is key to ensuring that optimal results are achieved in the fewest iterations, without getting stuck in local minima or maxima. Advanced Bayesian optimization include customized acquisition functions which have been used in autonomous electrochemistry studies for multi-objective and cost-aware approaches [23,43].

Autonomous experiments enable efficient performance optimization campaigns that would otherwise be prohibitively time-consuming. For example, Bayesian optimization has been applied for the development of solvent-electrolyte mixtures, adjusting the identity and amount of different co-solvents to improve critical metrics such as electrolyte conductivity, viscosity, and solubility [21,74,75]. A topic of interest has been the development of multi-metal heterogenous electrocatalysts. Notably, Kodera et al. performed fully autonomous optimization of Co−Mn−Fe−Ni composite-oxide electrocatalysts for seawater electrolysis, using Bayesian optimization to guide automated navigation of the possible quaternary phase diagram [18]. They performed 4 iterations of autonomous experiments, where 10 points in the first iteration were chosen at random, and the 10 points in each following iteration were suggested by Bayesian optimization. Figure 4a shows how the campaign progressed, with the target metric (current density) continuously improving after each iteration, finally settling around an optimum.

Figure 4



Examples of fully autonomous electrochemical experiments. (**a**) Bayesian optimization of Co–Mn–Fe–Ni composite-oxide electrocatalysts for seawater electrolysis, showing improved current densities as the experiment progress. Figure reproduced with permission from Kodera et al. [18] Copyright 2023, Royal Society of Chemistry. (**b**) Mult-objective Bayesian optimization (MOBO) of Co–Mn–Sb–Sn–Ti oxide electrocatalysts for the oxygen evolution reaction, simultaneously improving stability and activity. Norm. hypervolume is a measure of optimization performance. Figure reproduced with permission from Jenewein et al. [43] Copyright 2024, Royal Society of Chemistry. (**c**) Trajectories of Bayesian optimization campaign for mechanistic assignment of cobalt tetraphenylporphyrin (CoTPP) reacting with an organohalide, n-BuBr, iteratively adjusting scan rate and organohalide concentration. Later iteration give increased certainty that the reaction of interest follows the EC mechanism. Figure reproduced with permission from Sheng et al. [14] Copyright 2024, Springer Nature. (**d**) Bayesian optimization to optimize catalyst and substrate concentrations and scan rate to obtain kinetically insightful voltammograms. Figure reproduced with permission from Pence et al. [30].

Jenewein et al. performed a similar study on Co–Mn–Sb–Sn–Ti oxide electrocatalysts for the oxygen evolution reaction, using a multi-objective Bayesian optimization approach that simultaneously optimized for both stability and activity [43]. Figure 4b shows the performance of the optimization campaign over 4 iterations, with the Bayesian optimization guided experiments dramatically outperforming random search.

Not all autonomous electrochemistry experiments aim to optimize a performance metric. Instead, Bayesian optimization has proven to be an effective tool for iteratively adjusting solution and electrochemical parameters to maximize mechanistic insight. Sheng et al. conducted, to the best of our knowledge, the first autonomous electrochemistry experiment of this kind [14]. Their study employed a pump-valve-based AEP, a mechanism-discerning ML model, and Bayesian optimization to uncover the mechanism and kinetic parameters of the reaction between cobalt tetraphenylporphyrin (CoTPP) and organohalide electrophiles. Figure 4c illustrates the trajectory of Bayesian optimization as it adjusts scan rate and oraganohalide (n-BuBr) concentrations to confirm that the reaction follows an EC mechanism, where the electrochemical reduction of CoTPP is followed by an irreversible chemical reaction with the organohalide. Similarly, Figure 4d highlights work by us where we used autonomous electrochemistry and Bayesian optimization to study the TEMPO-catalyzed oxidation of alcohols, an example of molecular elecrocatalysis [30]. We used CV descriptors based on the kinetic zone diagram, a fundamental electrocatalysis concept, to iteratively adjust catalyst concentration, substrate concentration, and scan rate to obtain kinetically insightful voltammograms.

Outlook

One of the most important aspects of automated experiments is reproducibility and transferability of experimental protocols, especially given the rampant reproducibility crisis [76,77]. The automated and repeatable nature of these experiments helps prevent human error and explore the robustness of a result through identical execution protocols. If two platforms have compatible hardware and software, automation enables the direct transfer of experimental procedures so that other researchers can better reproduce reported results. Towards this end, there is a demand for platforms that are accessible and easy to construct using open-source hardware and software [78]. Many of the platforms highlighted in this article provide extensive supplemental information to help construct the platform, including software, circuit schematics, CAD files, bill of materials, and construction guides [15,16,24,31,67]. Often, the electrochemical instrumentation (i.e., the potentiostat) used in an AEP can be one of the most expensive parts. To reduce the cost needed for purchasing commercial potentiostats, there are a wide variety of affordably "do-it-yourself" potentiostats available in the literature [79].

We believe that the field of automated electrochemistry is just now beginning to truly flourish, with most of the notable works in the field occurring over the past two years. As automation and ML are embraced by the electrochemistry community, we expect to see AEPs applied to a wide variety of problems in energy, sustainability, and health. We are optimistic about the role automation and its integration with emerging artificial intelligence tools will play in accelerating discovery and unlocking the transformative potential of electrochemistry.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was funded by the Beckman Institute for Advanced Science and Technology Graduate Fellows Program with support from the Arnold and Mabel Beckman Foundation. Funding for this work was provided in part (support for G.H.) by the Robert C. and Carolyn J. Springborn Endowment.

Data availability

No data was used for the research described in the article.

References

Papers of particular interest, published within the period of review, have been highlighted as:

* of special interest
** of outstanding interest

1. Cabana J, Alaan T, Crabtree GW, Hatzell MC, Manthiram K, Steingart DA, Zenyuk I, Jiao F, Vojvodic A, Yang JY, Balsara NP, Persson KA, Siegel DJ, Haynes CL, Mauzeroll J, Shen M, Venton BJ, Balke N, Rodríguez-López J, Rolison DR, Shahbazian-Yassar R, Srinivasan V, Chaudhuri S, Couet A, Hattrick-Simpers J: **NGenE 2021: electrochemistry is everywhere**. *ACS Energy Lett* 2022, **7**:368–374, https://doi.org/10.1021/acsenergylett.1c02608.

2. Levin T, Bistline J, Sioshansi R, Cole WJ, Kwon J, Burger SP, Crabtree GW, Jenkins JD, O'Neil R, Korpås M, Wogrin S, Hobbs BF, Rosner R, Srinivasan V, Botterud A: **Energy storage solutions to decarbonize electricity through enhanced capacity expansion modelling**. *Nat Energy* 2023, **8**:1199–1208, https://doi.org/10.1038/s41560-023-01340-6.

3. Vikram A, Brudnak K, Zahid A, Shim M, A. Kenis PJ: **Accelerated screening of colloidal nanocrystals using artificial neural network-assisted autonomous flow reactor Technology**. *Nanoscale* 2021, **13**:17028–17039, https://doi.org/10.1039/D1NR05497J.

4. Steiner S, Wolf J, Glatzel S, Andreou A, Granda JM, Keenan G, Hinkley T, Aragon-Camarasa G, Kitson PJ, Angelone D, Cronin L: **Organic synthesis in a modular robotic system driven by a chemical programming language**. *Science* 2019, **363**, eaav2211, https://doi.org/10.1126/science.aav2211.

5. Strieth-Kalthoff F, Hao H, Rathore V, Derasp J, Gaudin T, Angello NH, Seifrid M, Trushina E, Guy M, Liu J, Tang X, Mamada M, Wang W, Tsagaantsooj T, Lavigne C, Pollice R, Wu TC, Hotta K, Bodo L, Li S, Haddadnia M, Wołos A, Roszak R, Ser CT, Bozal-Ginesta C, Hickman RJ, Vestfrid J, Aguilar-Granda A, Klimareva EL, Sigerson RC, Hou W, Gahler D, Lach S, Warzybok A, Borodin O, Rohrbach S, Sanchez-Lengeling B, Adachi C, Grzybowski BA, Cronin L, Hein JE, Burke MD, Aspuru-Guzik A: **Delocalized, asynchronous, closed-loop discovery of organic laser emitters**. *Science* 2024, **384**, eadk9227, https://doi.org/10.1126/science.adk9227.

6. Xu RHJ, Keating LP, Vikram A, Shim M, Kenis PJA: **Understanding hot injection quantum dot synthesis outcomes using automated high-throughput experiment platforms and machine learning**. *Chem Mater* 2024, **36**:1513–1525, https://doi.org/10.1021/acs.chemmater.3c02751.

7. Boiko DA, MacKnight R, Kline B, Gomes G: **Autonomous chemical research with large language models**. *Nature* 2023, **624**:570–578, https://doi.org/10.1038/s41586-023-06792-0.

8. Abolhasani M, Kumacheva E: **The rise of self-driving labs in chemical and materials sciences**. *Nat. Synth* 2023, **2**:483–492, https://doi.org/10.1038/s44160-022-00231-0.

9. Cullen LF, Brindle MP, Papariello GJ: **A system for automated polarographic analysis**. *J Pharmaceut Sci* 1973, **62**:1708–1713, https://doi.org/10.1002/jps.2600621031.

10. Wills AG, Charvet S, Battilocchio C, Scarborough CC, Wheelhouse KMP, Poole DL, Carson N, Vantourout JC: **High-throughput electrochemistry: state of the art, challenges, and perspective**. *Org Process Res Dev* 2021, **25**:2587–2600, https://doi.org/10.1021/acs.oprd.1c00167.

11. Xu X, Valavanis D, Ciocci P, Confederat S, Marcuccio F, Lemineur J-F, Actis P, Kanoufi F, Unwin PR: **The new era of high-throughput nanoelectrochemistry**. *Anal Chem* 2023, **95**:319–356, https://doi.org/10.1021/acs.analchem.2c05105.

12. Fung AWS, Knauer MJ, Blasutig IM, Colantonio DA, Kulasingam V: **Evaluation of electrochemiluminescence immunoassays for immunosuppressive drugs on the roche cobas E411 analyzer**. *F1000Research December* 2017, **12**, https://doi.org/10.12688/f1000research.12775.2.

13. Bell NL, Boser F, Bubliauskas A, Willcox DR, Luna VS, Cronin L: **Autonomous execution of highly reactive chemical transformations in the schlenkputer**. *Nat Chem Eng* 2024, **1**: 180–189, https://doi.org/10.1038/s44286-023-00024-y.

14. ** Sheng H, Sun J, Rodríguez O, Hoar BB, Zhang W, Xiang D, Tang T, Hazra A, Min DS, Doyle AG, Sigman MS, Costentin C, Gu Q, Rodríguez-López J, Liu C: **Autonomous closed-loop mechanistic investigation of molecular electrochemistry via automation**. *Nat Commun* 2024, **15**:2781, https://doi.org/10.1038/s41467-024-47210-x.

Sheng et al. demonstrated one of the first examples of an autonomous electrochemistry platform designed for mechanistic studies. This platform integrates a deep learning classifier, Bayesian optimization, and automated electrochemistry for mechanistic identification and kinetic analysis.

15. * Oh I, Pence MA, Lukhanin NG, Rodríguez O, Schroeder CM, Rodríguez-López J: **The Electrolab: an open-source, modular platform for automated characterization of redox-active electrolytes**. *Device* 2023, **1**, 100103, https://doi.org/10.1016/j.device.2023.100103.

Oh et al. present an automated electrochemistry platform that utilizes a low-cost modified 3D printer as a gantry.

16. Pence MA, Hazen G, Rodríguez-López J: **An automated electrochemistry platform for studying pH-dependent molecular electrocatalysis**. *Digital Discovery* 2024, https://doi.org/10.1039/D4DD00186A.

17. Dieckhöfer S, Schuhmann W, Ventosa E: **Accelerated electrochemical investigation of Li plating efficiency as key parameter for Li metal batteries utilizing a scanning droplet cell**. *Chemelectrochem* 2021, **8**:3143–3149, https://doi.org/10.1002/celc.202100733.

18. ** Kodera M, Sayama K: **An automatic robot system for machine learning–assisted high-throughput screening of composite electrocatalysts**. *Digital Discovery* 2023, **2**:1683–1687, https://doi.org/10.1039/D3DD00116D.

Kodera et al. use a dual robotic arm platform to perform studies of mixed-metal oxide electrocatalysts, enabling sophisticated multi-step and multi-instrument operations.

19. * Quinn H, A. Robben G, Zheng Z, L. Gardner A, Werner J G, Brown A, Panda K: **A self-driving lab for studying electrodeposited polymer films**. *Mater Horiz* 2024, **11**:5331–5340, https://doi.org/10.1039/D4MH00797B.

The work by Quinn et al. is a unique example of automated multi-modal analysis, measuring the color change of electrochromic devices during the application of potential. This platform was employed to test the electrochromic performance of electrodeposited materials in a fully closed-loop workflow driven by Bayesian optimization.

20. Darvish K, Skreta M, Zhao Y, Yoshikawa N, Som S, Bogdanovic M, Cao Y, Hao H, Xu H, Aspuru-Guzik A, Garg A, Shkurti F: **ORGANA: a robotic assistant for automated Chemistry experimentation and characterization**. *Matter* 2025, **8**, https://doi.org/10.1016/j.matt.2024.10.015.

21. Dave A, Mitchell J, Burke S, Lin H, Whitacre J, Viswanathan V: **Autonomous optimization of non-aqueous Li-ion battery electrolytes via robotic experimentation and machine learning coupling**. *Nat Commun* 2022, **13**:5454, https://doi.org/10.1038/s41467-022-32938-1.

22. Zahrt AF, Mo Y, Nandiwale KY, Shprints R, Heid E, Jensen KF: **Machine-learning-guided discovery of electrochemical reactions**. *J Am Chem Soc* 2022, **144**:22599–22610, https://doi.org/10.1021/jacs.2c08997.

23. Liang R, Zheng S, Wang K, Yuan Z: **Cost-aware bayesian optimization for self-driven condition screening of flow electrosynthesis**. *ACS Electrochem* 2024, https://doi.org/10.1021/acselectrochem.4c00078.

24. Mo Y, Rughoobur G, Nambiar AMK, Zhang K, Jensen KF: **A multifunctional microfluidic platform for high-throughput experimentation of electroorganic Chemistry**. *Angew Chem Int Ed Engl* 2020, **59**:20890–20894, https://doi.org/10.1002/anie.202009819.

25. Rial-Rodríguez E, Williams JD, Cantillo D, Fuchß T, Sommer A, Eggenweiler H-M, Kappe CO, Laudadio G: **An automated electrochemical flow platform to accelerate library synthesis and reaction optimization**. *Angew Chem* 2024, **136**, e202412045, https://doi.org/10.1002/ange.202412045.

26. Gao RZ, Hébert M, Huissoon J, Ren CL: **μPump: An open-source pressure pump for precision fluid handling in microfluidics**. *HardwareX* 2020, **7**, e00096, https://doi.org/10.1016/j.ohx.2020.e00096.

27. Davis JJ, Padalino M, Kaplitz AS, Murray G, Foster SW, Maturano J, Grinias JP: **Utility of low-cost, miniaturized peristaltic and venturi pumps in droplet microfluidics**. *Anal Chim Acta* 2021, **1151**, 338230, https://doi.org/10.1016/j.aca.2021.338230.

28. Baas S, Saggiomo V: **Ender3 3D printer kit transformed into open, programmable syringe pump set**. *HardwareX* 2021, **10**, e00219, https://doi.org/10.1016/j.ohx.2021.e00219.

29. Lake JR, Heyde KC, Ruder WC: **Low-cost feedback-controlled syringe pressure pumps for microfluidics applications**. *PLoS One* 2017, **12**, e0175089, https://doi.org/10.1371/journal.pone.0175089.

30. ** Pence M, Hazen G, Rodríguez-López J: **Closed-loop navigation of a kinetic zone diagram for redox-mediated electrocatalysis using bayesian optimization, a digital twin, and automated electrochemistry**. *Anal Chem* 2025, https://doi.org/10.1021/acs.analchem.5c00099.

Pence et al. used the eLab platform to conduct autonomous studies of molecular electrocatalysis. Conceptual frameworks from electrocatalysis theory were applied to develop a Bayesian optimization workflow, which identified conditions that produced kinetically informative voltammograms.

31. Abed J, Bai Y, Persaud D, Kim J, Witt J, Hattrick-Simpers J, Sargent H, Ampere E: **Automated modular platform for expedited and reproducible electrochemical testing**. *Digital Discovery* 2024, https://doi.org/10.1039/D4DD00203B.

32. Liang S, Kinghorn AB, Voliotis M, Prague JK, Veldhuis JD, Tsaneva-Atanasova K, McArdle CA, Li RHW, Cass AEG, Dhillo WS, Tanner JA: **Measuring luteinising hormone pulsatility with a robotic aptamer-enabled electrochemical reader**. *Nat Commun* 2019, **10**:852, https://doi.org/10.1038/s41467-019-08799-6.

33. Haššo M, Kudr J, Zítka J, Šílený J, Švec P, Švorc Ľ, Zítka O: **Proving the automatic benchtop electrochemical station for the development of dopamine and paracetamol sensors**. *Microchim Acta* 2024, **191**:408, https://doi.org/10.1007/s00604-024-06454-6.

34. Erichsen T, Reiter S, Ryabova V, Bonsen EM, Schuhmann W, Märkle W, Tittel C, Jung G, Speiser B: **Combinatorial microelectrochemistry: development and evaluation of an electrochemical robotic system**. *Rev Sci Instrum* 2005, **76**, 062204, https://doi.org/10.1063/1.1906106.

35. Briehn CA, Schiedel M-S, Bonsen EM, Schuhmann W, Bäuerle P: **Single-compound libraries of organic materials: from the combinatorial synthesis of conjugated oligomers to structure–property relationships**. *Angew Chem Int Ed* 2001, **40**:4680–4683, https://doi.org/10.1002/1521-3773(20011217)40:24<4680::AID-ANIE4680>3.0.CO;2-X.

36. Intarakamhang S, Schulte A: **Automated electrochemical free radical scavenger screening in dietary samples**. *Anal Chem* 2012, **84**:6767–6774, https://doi.org/10.1021/ac301292c.

37. Godfrey D, Bannock JH, Kuzmina O, Welton T, Albrecht T: **A robotic platform for high-throughput electrochemical analysis of chalcopyrite leaching**. *Green Chem* 2016, **18**: 1930–1937, https://doi.org/10.1039/C5GC02306H.

38. Zeng JS, Padia V, Chen GY, Maalouf JH, Limaye AM, Liu AH, Yusov MA, Hunter IW, Manthiram K: **Nonidealities in CO2 electroreduction mechanisms revealed by automation-assisted kinetic analysis**. *ACS Cent Sci* 2024, **10**:1348–1356, https://doi.org/10.1021/acscentsci.3c01295.

39. Sliozberg K, Schäfer D, Erichsen T, Meyer R, Khare C, Ludwig A, Schuhmann W: **High-throughput screening of thin-film semiconductor material libraries I: system development and case study for TiWO**. *ChemSusChem* 2015, **8**:1270–1278, https://doi.org/10.1002/cssc.201402917.

40. Ebejer N, Güell AG, Lai SCS, McKelvey K, Snowden ME, Unwin PR: **Scanning electrochemical cell microscopy: a versatile technique for nanoscale electrochemistry and functional imaging**. *Annual Review of Analytical Chemistry 2013* 2013, **6**:329–351, https://doi.org/10.1146/annurev-anchem-062012-092650.

41. Sur D, Joress H, Hattrick-Simpers J, Scully JR: **A high throughput aqueous passivation testing methodology for compositionally complex alloys using a scanning droplet cell**. *J Electrochem Soc* 2023, **170**, 081507, https://doi.org/10.1149/1945-7111/aceeb8.

42. Lai Z, Zou Y, Zhao Z, Huang F, Liu P, Lai T, Jin Y: **An automated test platform for high-throughput micro-electrochemical characterization of metallic materials and its application on a Fe–Cr–Ni combinatorial materials chip**. *J Electrochem Soc* 2021, **168**, 091501, https://doi.org/10.1149/1945-7111/ac24bc.

43. ** Jenewein KJ, Torresi L, Haghmoradi N, Kormányos A, Friederich P, Cherevko S: **Navigating the unknown with AI: multiobjective bayesian optimization of non-noble acidic OER catalysts**. *J Mater Chem A* 2024, **12**:3072–3083, https://doi.org/10.1039/D3TA06651G.
Jenewein et al. employed a scanning droplet cell approach for multi-objective optimization of oxygen evolution electrocatalysts, simultaneously optimizing both catalyst activity and stability.

44. Garcia G, Ventosa E, Schuhmann W: **Complete prevention of dendrite formation in Zn metal anodes by means of pulsed charging protocols**. *ACS Appl Mater Interfaces* 2017, **9**:18691–18698, https://doi.org/10.1021/acsami.7b01705.

45. Gregoire JM, Xiang C, Liu X, Marcin M, Jin J: **Scanning droplet cell for high throughput electrochemical and photoelectrochemical measurements**. *Rev Sci Instrum* 2013, **84**, 024102, https://doi.org/10.1063/1.4790419.

46. Muñoz-Torrero D, Santos CS, García-Quismondo E, Dieckhöfer S, Erichsen T, Palma J, Schuhmann W, Ventosa E: **The redox mediated – scanning droplet cell system for evaluation of the solid electrolyte interphase in Li-ion batteries**. *RSC Adv* 2023, **13**:15521–15530, https://doi.org/10.1039/D3RA00631J.

47. Sanin A, Stein HS: **Exploring reproducible nonaqueous scanning droplet cell electrochemistry in model battery chemistries**. *Chem Mater* 2024, **36**:3536–3545, https://doi.org/10.1021/acs.chemmater.3c01768.

48. Kollender JP, Voith M, Schneiderbauer S, Mardare AI, Hassel AW: **Highly customisable scanning droplet cell microscopes using 3D-printing**. *J Electroanal Chem* 2015, **740**:53–60, https://doi.org/10.1016/j.jelechem.2014.12.043.

49. Duke R, Mahmoudi S, Preet Kaur A, Bhat V, C. Dingle I, C. Stumme N, K. Shaw S, Eaton D, Vego A, Risko C: **ExpFlow: a graphical user interface for automated reproducible electrochemistry**. *Digital Discovery* 2024, **3**:163–172, https://doi.org/10.1039/D3DD00156C.

50. Fatehi E, Thadani M, Birsan G, Black RW: **A critical evaluation of a self-driving laboratory for the optimization of electrodeposited earth-abundant mixed-metal oxide catalysts for the oxygen evolution reaction (OER)**. *arXiv* 2023, https://doi.org/10.48550/arXiv.2305.12541. November 13.

51. * Yoshikawa N, Akkoc GD, Pablo-García S, Cao Y, Hao H, Aspuru-Guzik A: **Does one need to polish electrodes in an eight pattern? Automation provides the answer**. *Digital Discovery* 2025, https://doi.org/10.1039/D4DD00323C.
The automated electrochemistry platform by Yoshikawa et al. used a robotic arm to perform oft-overlooked electrochemical operations such as electrode polishing.

52. Alden SE, Zhang L, Wang Y, Lavrik NV, Thorgaard SN, Baker LA: **High-throughput single-entity electrochemistry with microelectrode arrays**. *Anal Chem* 2024, **96**:9177–9184, https://doi.org/10.1021/acs.analchem.4c01092.

53. Takada K, Fujimoto K, Sasaki T, Watanabe M: **Combinatorial electrode array for high-throughput evaluation of combinatorial library for electrode materials**. *Appl Surf Sci* 2004, **223**:210–213, https://doi.org/10.1016/S0169-4332(03)00924-3.

54. Sullivan MG, Utomo H, Fagan PJ, Ward MD: **Automated electrochemical analysis with combinatorial electrode arrays**. *Anal Chem* 1999, **71**:4369–4375, https://doi.org/10.1021/ac990331y.

55. Pence MA, Rodríguez O, Lukhanin NG, Schroeder CM, Rodríguez-López J: **Automated measurement of electrogenerated redox species degradation using multiplexed interdigitated electrode arrays**. *ACS Meas. Sci. Au* 2023, **3**:62–72, https://doi.org/10.1021/acsmeasuresciau.2c00054.

56. Gaddam R, Wang Z, Li Y, Harris LC, Pence MA, Guerrero ER, Kenis PJA, Gewirth AA, Rodríguez-López J: **Identifying reactive trends in glycerol electro-oxidation using an automated screening approach: 28 ways to electrodeposit an Au electrocatalyst**. *ACS Catal* 2024:639–652, https://doi.org/10.1021/acscatal.4c04190.

57. Chang Y-C, Arnould B, Heemstra JM, Moeller KD: **Developing microelectrode arrays for the point-of-care multiplex detection of metabolites**. *Anal Chem* 2024, **96**:14571–14580, https://doi.org/10.1021/acs.analchem.4c02978.

58. Clark RB, Glasscott MW, Verber MD, DeMartino JC, Netchaev A, Ray JD, Brown EW, Alberts E, Fernando PUAI, Moores LC, Dick JE: **A generalized potentiostat adaptor for multiplexed electroanalysis**. *Anal Chem* 2021, **93**:7381–7387, https://doi.org/10.1021/acs.analchem.0c05299.

59. Rein J, Annand JR, Wismer MK, Fu J, Siu JC, Klapars A, Strotman NA, Kalyani D, Lehnherr D, Lin S: **Unlocking the potential of high-throughput experimentation for electrochemistry with a standardized microscale reactor**. *ACS Cent Sci* 2021, **7**:1347–1355, https://doi.org/10.1021/acscentsci.1c00328.

60. Gerroll BHR, Kulesa KM, Ault CA, Baker LA: **Legion: an instrument for high-throughput electrochemistry**. *ACS Meas. Sci. Au* 2023, https://doi.org/10.1021/acsmeasuresciau.3c00022.

61. Kulesa KM, Hirtzel EA, Nguyen VT, Freitas DP, Edwards ME, Yan X, Baker LA: **Interfacing high-throughput electrosynthesis and mass spectrometric analysis of azines**. *Anal Chem* 2024, **96**:8249–8253, https://doi.org/10.1021/acs.analchem.4c01110.

62. Personick ML, Jallow AA, Halford GC, Baker LA: **Nanomaterials synthesis discovery via parallel electrochemical deposition**. *Chem Mater* 2024, **36**:3034–3041, https://doi.org/10.1021/acs.chemmater.4c00318.

63. * Górski B, Rein J, Norris S, Ji Y, McEuen PL, Lin S: **Light-harvesting microelectronic devices for wireless electrosynthesis**. *Nature* 2025, **637**:354–361, https://doi.org/10.1038/s41586-024-08373-1.
Górski et al. developed a microfabricated photovoltaic-based device for high-throughput electrosynthesis screening. These devices can be placed in a 96-well plate, each functioning as an independent constant current source to drive electrochemical reactions.

64. Seifrid M, Pollice R, Aguilar-Granda A, Morgan Chan Z, Hotta K, Ser CT, Vestfrid J, Wu TC, Aspuru-Guzik A: **Autonomous chemical experiments: challenges and perspectives on establishing a self-driving lab**. *Acc Chem Res* 2022, **55**:2454–2466, https://doi.org/10.1021/acs.accounts.2c00220.

65. * Laws K, Tze-Kiat Ng M, Sharma A, Jiang Y, Hammer AJS, Cronin L: **An autonomous electrochemical discovery robot that utilises probabilistic algorithms: probing the redox behaviour of inorganic materials**. *Chemelectrochem* 2024, **11**, e202300532, https://doi.org/10.1002/celc.202300532.
The work by Laws et al. integrates an automated synthesis robot with automated electrochemical characterization to study polyoxometalates. They demonstrate a fully closed-loop design–make–test–analyze cycle, optimizing voltammetric capacitance.

66. Salley DS, Keenan GA, Long D-L, Bell NL, Cronin L: **A modular programmable inorganic cluster discovery robot for the discovery and synthesis of polyoxometalates**. *ACS Cent Sci* 2020, **6**:1587–1593, https://doi.org/10.1021/acscentsci.0c00415.

67. Pablo-García S, García Á, Akkoc GD, Sim M, Cao Y, Somers M, Hattrick C, Yoshikawa N, Dworschak D, Hao H, Aspuru-Guzik A: **An affordable platform for automated synthesis and electrochemical characterization**. *Device* 2024, 100567, https://doi.org/10.1016/j.device.2024.100567.

68. Szymanski NJ, Rendy B, Fei Y, Kumar RE, He T, Milsted D, McDermott MJ, Gallant M, Cubuk ED, Merchant A, Kim H, Jain A,

Bartel CJ, Persson K, Zeng Y, Ceder G: **An autonomous laboratory for the accelerated synthesis of novel materials**. *Nature* 2023, **624**:86–91, https://doi.org/10.1038/s41586-023-06734-w.

69. Angello NH, Friday DM, Hwang C, Yi S, Cheng AH, Torres-Flores TC, Jira ER, Wang W, Aspuru-Guzik A, Burke MD, Schroeder CM, Diao Y, Jackson NE: **Closed-loop transfer enables artificial intelligence to yield chemical knowledge**. *Nature* 2024, **633**:351–358, https://doi.org/10.1038/s41586-024-07892-1.

70. Minteer SD, Baran P: **Electrifying synthesis: recent advances in the methods, materials, and techniques for organic electrosynthesis**. *Acc Chem Res* 2020, **53**:545–546, https://doi.org/10.1021/acs.accounts.0c00049.

71. Tom G, Schmid SP, Baird SG, Cao Y, Darvish K, Hao H, Lo S, Pablo-García S, Rajaonson EM, Skreta M, Yoshikawa N, Corapi S, Akkoc GD, Strieth-Kalthoff F, Seifrid M, Aspuru-Guzik A: **Self-driving laboratories for Chemistry and materials science**. *Chem Rev* 2024, **124**:9633–9732, https://doi.org/10.1021/acs.chemrev.4c00055.

72. * Wu Y, Walsh A, Ganose M: **A. Race to the bottom: bayesian optimisation for chemical problems**. *Digital Discovery* 2024, **3**: 1086–1100, https://doi.org/10.1039/D3DD00234A.
Wu et al. give a comprehensive and chemist-friendly introduction to Bayesian optimization.

73. Frazier PI: **A tutorial on bayesian optimization**. *arXiv* 2018, https://doi.org/10.48550/arXiv.1807.02811. July 8.

74. Dave A, Mitchell J, Kandasamy K, Wang H, Burke S, Paria B, Póczos B, Whitacre J, Viswanathan V: **Autonomous discovery of battery electrolytes with robotic experimentation and machine learning**. *Cell Reports Physical Science* 2020, **1**, 100264, https://doi.org/10.1016/j.xcrp.2020.100264.

75. * Noh J, Doan HA, Job H, Robertson LA, Zhang L, Assary RS, Mueller K, Murugesan V, Liang Y: **An integrated high-throughput robotic platform and active learning approach for accelerated discovery of optimal electrolyte formulations**. *Nat Commun* 2024, **15**:2757, https://doi.org/10.1038/s41467-024-47070-5.
This work by Noh et al. employs a Bayesian workflow to study electrolyte solubility, a key obstacle in the practical deployment of redox-flow batteries. The automated platform dispenses both solid samples and solvents to prepare solutions for characterization.

76. Fanelli D: **Is science really facing a reproducibility crisis, and do we need it to?** *Proc Natl Acad Sci USA* 2018, **115**: 2628–2631, https://doi.org/10.1073/pnas.1708272114.

77. Baker M: **1,500 scientists lift the lid on reproducibility**. *Nature* 2016, **533**:452–454, https://doi.org/10.1038/533452a.

78. * Rodríguez O, Pence MA, Rodríguez-López J: **Hard potato: a Python library to control commercial potentiostats and to automate electrochemical experiments**. *Anal Chem* 2023, **95**:4840–4845, https://doi.org/10.1021/acs.analchem.2c04862.
Rodríguez et al. present an open-source Python library designed for the automated control and operation of electrochemical instrumentation.

79. von Zuben TW, Salles AG, Bonacin JA: **Low-cost open-source potentiostats: a comprehensive review of diy solutions and fundamental concepts of electronics and its integration with electrochemistry**. *Electrochim Acta* 2024, **498**, 144619, https://doi.org/10.1016/j.electacta.2024.144619.